

SEMESTER END EXAMINATIONS - APRIL 2023

Program	: B.E :- Computer Science and Engineering	Semester	: III
Course Name	: Digital Design and Computer Organization	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Using Karnaugh Map, Simplify the given Boolean expression: $f(x_1, x_2, x_3, x_4) = \sum m(4, 6, 8, 10, 11, 12, 15) + D(3, 5, 7, 9)$ and implement the same using NAND Gate. CO1 (08)
 - What is Multiplexer? Implement the following function using 8:1 MUX $f(A, B, C, D) = \sum m(1, 2, 5, 6, 9, 12)$ CO1 (06)
 - Briefly discuss about Priority Encoder. CO1 (06)
- Using Tabular method, find the essential prime implicants. $f(x_1, \dots, x_4) = \sum m(0, 2, 4, 5, 7, 8, 9, 15)$ CO1 (08)
 - Design BCD to 7-Segment display code convertor. CO1 (08)
 - Derive the minimized form of the following Boolean expressions: CO1 (04)
 - $(X+Y)(X+Y') + ((XY') + X')'$
 - $(A'B'C' + A'BC' + A'BC + ABC')$

UNIT - II

- Draw the logic diagram of the D flip-flop and explain the working with a suitable timing diagram. CO2 (06)
 - Design a 3-bit Asynchronous down counter using T-FF and draw the timing diagram for it. CO2 (06)
 - Derive the characteristic equation for SR-FF, D-FF, JK-FF and T-FF. CO2 (08)
- With a neat diagram explain the Parallel Access shift register. CO2 (06)
 - Design mod 3 self-correcting synchronous up counter using D-FF. CO2 (06)
 - Differentiate between Synchronous and Asynchronous sequential counters. CO2 (08)
 - Design a random counter for the sequence 0-3-1-2-0 using T-FF.

UNIT - III

- Discuss the parameters that determines the performance of a computer. CO3 (06)
 - Illustrate reasons for arithmetic overflow that occur during arithmetic operations also identify the overflow in the following operations: CO3 (07)

$$\begin{array}{r}
 0111 \quad 1101 \quad 1001 \\
 + 1101 \quad + 0111 \quad - 1011 \\
 \hline
 \end{array}$$

- Explain the following with an example: CO3 (07)
 - Index addressing mode
 - PC relative addressing mode
 - Auto increment addressing mode

6. a) Explain different formats of encoding machine instruction for different Addressing modes. CO3 (06)
b) Write an assembly level program to find the factorial of a number using subroutine. CO3 (06)
c) Explain the connection between the Processor and the Memory with a neat diagram. CO3 (08)

UNIT- IV

7. a) Write the control sequence for the instruction Add (R3), R1 for single bus organization. CO4 (05)
b) Justify the statement with respect to the pipeline technique: "If there are no stalls, the speedup is equal to the number of pipeline stages". CO4 (05)
c) Write Booth Multiplication Algorithm. Apply Booth Multiplication Algorithm to multiply the signed number -5 and 4. CO4 (10)
8. a) Draw and explain multiple bus organization along with its advantages. CO4 (08)
b) With an example show how data forwarding technique is used to resolve data hazard. CO4 (08)
c) Represent -307.1875 in single and double precision IEEE floating point numbers. CO4 (04)

UNIT - V

9. a) With figure, explain about Direct mapping Cache memory. CO5 (05)
b) Illustrate the hardware required to connect an I/O device to the bus with neat diagram and also explain registers in keyboard and display interface. CO5 (10)
c) What is an interrupt? With example illustrate the concepts of interrupts. CO5 (05)
10. a) Describe connection of the memory to the processor with neat diagram. CO5 (05)
b) With a neat diagram explain the processor connection to the keyboard and printer. CO5 (05)
c) Discuss enabling and disabling interrupts. Also explain how to handle multiple devices. CO5 (10)
